Lithological Unit Classification Based on Geological Knowledge-Guided Deep Learning Framework for Optical Stereo Mapping Satellite Imagery

Gaodian Zhou, Weitao Chen, *Member, IEEE*, Xuwen Qin, Jun Li, *Fellow, IEEE*, and Lizhe Wang, *Fellow, IEEE*

***Abstract*— Lithological unit classification (LUC) refers to the classification of different types of rocks within an area, and it has been widely used in many fields, such as resource surveys and infrastructure planning. However, traditional field surveys require a lot of resources and time. Since remote sensing technology can rapidly acquire information without regional limitations, many researchers have focused on classifying lithological units with remote sensing images. However, in an area covered by vegetation, the beneficial information directly provided by remote sensing images is limited. Moreover, lithological interpretation often requires geological prior knowledge for guidance, which cannot be provided by remote sensing images. Thus, this study designed a dual-branch deep learning model to extract geological prior knowledge from geological information, and improve the accuracy of LUC. In the process of feature transmission of the model, a Dense Attention residual-atrous spatial pyramid pooling (DA-ASPP) module was proposed to maximize the preservation of lithological units' features. The DA-ASPP integrates the idea of dense connection into ASPP for multiscale object feature preservation and adds residual structure into the channel attention mechanism to screen out the representative features of lithological units. The study area was located in southeastern Hubei Province, China, with seven categories of lithological units. A total of seven deep learning networks were compared. The proposed method achieved a mean intersection over union (IOU) of 44.61% with a Macro-F1 of 56.54%, which were better than those of comparison models. Moreover, the visualization results demonstrated the superiority of the proposed model in LUC.**

***Index Terms*— Dense attention residual, dual branches, lithological unit classification (LUC), prior knowledge, remote sensing.**

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I. Introduction

LITHOLOGICAL unit classification (LUC) is important for resource reconnaissance, infrastructure planning, and landslide/environmental governance [1], [2], [3], [4], [5], [6], [7], [8], [9]. Traditional field surveys remain the main mode of geological data collection and lithological mapping. However, field surveys are difficult and time-consuming in deserts, plateaus, polar, and other inaccessible areas.

Remote sensing has been used for Earth observations since the advent of imaging spectrometer systems [10], [11], [12], [13]. Compared to field surveys, it is a cost-effective and efficient method for collecting information from remote areas with limited accessibility [14], [15], [16]. Hence, some researchers have designed various models to extract the features of remote sensing image data and interpret the lithology [17], [18], [19], [20]. These models can be divided into handcraft-feature- and depth-feature-based models. For handcrafted feature extraction research, a model is constructed to extract specific features, such as spectral and texture features; machine learning methods are used for lithologic classification [4], [17], [18], [19]. For depth feature extraction research, multilayer neural networks are designed to learn high-level features and classify lithological units [21], [22]. With the development of deep learning in computer vision, researchers have focused on depth-feature-based models.

A. Handcraft Feature Extraction

Handcraft feature of lithological units can be divided into two types, viz., spectral features and textual features. Spectral features are the essential features that distinguish different lithological units. Textual features are the spatial relationship of lithological units, which does not depend on the color and brightness of objects and can reflect their spatial arrangement. Commonly used spectral features include mineral spectral reflectance, spectral shape, and vegetation index [23], [24]. Whereas textual features include digital elevation model (DEM), shade, slope, and aspect [25], [26], [27], [28]. Hence, some researchers have attempted to use these characteristics for the lithologic identification of remote sensing images.

Rowan et al. [25] analyzed the spectral shape, albedo, and specific spectral features to classify ultrabasic rocks. The result indicated the importance of analyzing specific spectral features

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for mapping lithologic units. Yu et al. [26] extracted seven types of features from the Advanced Space-borne Thermal Emission and Reflection Radiometer (ASTER), DEM, and aeromagnetic data. Then, a support vector machine (SVM) algorithm was applied to classify the lithology in northwestern India. Wang et al. [27] extracted textural features from ASTER, adopted random forest metric learning to integrate textual features and geochemical data, and classified seven lithological target units in the northeastern part of the Himalayas, China. The result demonstrated the effectiveness of the hybrid method and multihandcraft features for geological exploration in inaccessible areas. Perez et al. [28] extracted rock color and textual features and selected 14 features to represent the information of rocks. Then, three SVM classifiers were applied to interpret different rocks. Bachri et al. [17] combined DEM geomorphometric attributes with the spectral features of Landsat 8 Operational Land Imager and classified 12 types of lithological units using SVM; the result indicated an overall classification accuracy of 85% which confirmed the importance of handcraft features.

The above research extracted features through mathematical derivation and transformation, which can represent the specific attribute information of high-resolution images. However, for remote sensing images, which contain a large amount of information, the expression ability of these features is limited, especially for high-dimensional information. Therefore, many researchers have applied deep learning to feature extraction to use remote sensing features fully.

B. Feature Extraction of Different Deep Learning Strategies

Convolutional neural networks (CNNs) are widely used in image classification and overcome a series of problems, such as heavy workload or limited representativeness caused by handcrafted feature extraction. Recently, many researchers have proposed models with different strategies to improve classification accuracy.

1) Multiscale Information: Different scales of feature maps can help the CNNs model classify objects with different pixel-wise dimensions [29], [30]; this can be extracted using multiple filter banks of different scales [31], [32]. Some researchers have proposed specific models to extract multiscale features. For example, the inception module [33], [34], [35], [36] used convolution filters of different sizes and then concentrated them to enrich the information of each layer. Spatial pyramid pooling network [37], pyramid scene parsing network (PSPNet) [38], and atrous spatial pyramid pooling (ASPP) [39], [40], [41] constructed pooling filters with different sizes and strides and then combined these features. High resolution network (HRNet) [42], [43], [44] repeatedly exchanged information on multiresolution subnetworks for multiscale feature fusion. Omni-scale network [45], [46] designed a residual block including multiple branches of convolution features. Each branch can extract features of a certain scale.

2) Attention Mechanism: The attention mechanism focuses on information more critical to the current task in input information to assign computing power to more important tasks [47], [48], [49], [50]. Computer vision has four main attention mechanisms: channel, spatial, self-, and hybrid attention [50], [51], [52]. Channel attention (e.g., squeeze and extraction network [53], efficient channel attention network [54], and frequency channel attention network [55]) generates an attention mask across the channel domain and uses it to select the important channels. Spatial attention (e.g., gather excite network [56] and point-wise spatial attention network [57]) generates attention masks across spatial domains and uses them to select important spatial regions. Self-attention (e.g. transformer [47], [58], ConvNet [59], and visual attention network [60]) compares and relates all input features of different positions with each other and computes a representation of the same feature. Hybrid attention fuses with and takes advantage of other attention mechanisms. Chen et al. [61] combined channel and self-attention models and proposed an overlapping cross-attention model. Li et al. [62] fused channel and spatial attention for image classification.

3) Contextual Information: Contextual data is information on the background surrounding other data points and can improve the understanding of an image scene [56], [63], [64]. The essence of acquiring contextual information expands the receptive field. Dilated convolution [65], [66], [67] expands the receptive field by inserting holes between the adjacent elements in the filter. Conditional random fields (CRFs) [39], [68], [69] construct an undirected probabilistic graph to extract contextual information. Graph convolution networks (GCNs) [70], [71], [72] obtain a wider range of features by building an adjacency matrix and changing the propagation path of features. Transformer [44], [58], [61] constructs multihead self-attention to extract global contextual information.

4) Module Fusion: Many researchers have integrated and modified the modules, proposing their networks to take advantage of each module. For example, Gu et al. [73] extracted multiscaled information and applied a transformer to learn semantically rich and spatially precise representations. Li et al. [74] proposed a contextual transformer block to capitalize on contextual information and strengthen the capacity for visual representations. Cao et al. [75] used a Swin transformer module [58] instead of a convolutional layer in UNet [76] for medical image segmentation.

Deep learning has been well-developed in computer vision. However, there is a lack of relevant research on LUC. Some studies have only applied the classical deep learning model to classify lithological units. For example, Wang et al. [15] applied a fully convolutional network (FCN) [77] to classify seven lithological units in the Cuonadong Dome, Tibet, China. Shirmard et al. [16] constructed a CNNs model with only two convolution layers and compared it with machine learning methods, providing the best performance for LUC. Han et al. [14] proposed an adaptive multisource data-fusion network to extract spatial and spectral information for the classification of lithological units.

Previous research has made progress in classifying lithologic units. However, in areas covered by vegetation, the information provided by remote sensing images is very limited, restricting the accuracy of LUC. In a study conducted

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by Han et al. [14] in 2022, problems caused by vegetation coverage were not considered, as almost all of their research areas were bare land. In traditional geological mapping, lithologic units in certain areas (especially inaccessible areas and areas with vegetation coverage) are inferred from geological background information. In other words, prior geological knowledge plays an important role in lithological unit identification [78]. However, to our knowledge, no research utilizes deep learning features and prior geological knowledge for LUC. Furthermore, in the process of lithologic feature extraction, with an increase in the number of model layers, some lithologic unit features are lost in feature propagation. This loss makes it difficult for the model to identify these lithologic units, especially small-scale ones.

C. Main Contributions of This Study

This study designed a dual-branch network to classify the lithological units to address these issues. One branch, the prior knowledge branch, was used to extract geological background information from small-scale geological lithological unit information. The other branch, the remote sensing branch, was applied to extract features from remote sensing imagery. Furthermore, an improved ASPP module was proposed, integrating dense connection into ASPP and adding a residual structure into the channel attention mechanism to extract multiscale features and screen out representative features of lithologic units. When merging prior knowledge and remote sensing features from these two branches, a convolution layer was applied to project them to the same dimension and fuse them with residual channel attention. Then, transposed convolution was used to decode the fused feature. A lithological unit semantic segmentation dataset in Tieshan District, Huangshi City, Hubei Province, China, was constructed to demonstrate the effectiveness of the proposed model. In addition, seven deep learning models were used to compare the performance of the proposed model, including UNet, PSPNet [52], Deeplab V3+, Attention UNet, UNet++, and some recent models, Swin UNet [75] and Bi-HRNet [79].

This study has two innovations. The first is using small-scale geological lithological unit information as prior geological knowledge to provide geological background information. The second innovation proposes a dual-branch deep learning model with an improved ASPP module for multiscale feature extraction and screening out the representative features of lithologic units.

II. Methods

The proposed dual-branch prior knowledge model (DPNet) comprises prior knowledge and remote sensing branches. In the feature extraction stage, both branches used ResNet-50 as the backbone to extract deep features and project the features of different branches to the same dimension. Then, the residual channel attention module was applied to screen out the important features of the lithologic units. Finally, the features of the two branches were fused and decoded using transposed convolution. Furthermore, an improved ASPP module (DA-ASPP, dense attention residual ASPP) was added to ResNet-50 to extract multiscale information and filter out effective channel features.

A. Geological Prior Knowledge

In geological surveys, the overall process is from small-scale to large-scale, and from coarse to fine. Large-scale geological surveys are based on the results of small-scale geological surveys. Intensive sampling and drilling make the survey results more accurate. Inspired by this, when classifying large-scale (1:50 000) lithological units, small-scale (1:200 000) lithological data can provide more detailed geological information. Furthermore, digital terrain models (DTMs), which are similar to DEMs, are useful in LUC. Hence, small-scale geological lithological unit information, remote sensing images, and DTM were merged as prior knowledge (see Fig. 1). According to the relationship between the spatial resolution and map scale [80], 30-m spatial resolution remote sensing imagery was used to adapt to the scale of 1:200 000 geological lithological unit information. Furthermore, a 2-m spatial resolution optical remote sensing image and 5-m spatial resolution DTM were merged to provide remote sensing information. Finally, prior knowledge and remote sensing information were applied to classify the 1:50 000 scale geological lithological units.

B. Structure of Proposed Network

The DPNet comprises four stages (see Fig. 2): feature extraction, feature projection, feature fusion, and upsampling. The input of the prior knowledge branch consisted of small-scale geological lithological unit information (1:200 000), optical remote sensing imagery (30 m), and DTM (30 m). And the input of remote sensing branch consisted of optical imagery (2.1 m) and DTM (2.1 m).

1) Feature Extraction: The feature extraction stage is mainly based on the residual network (ResNet) [81], it is widely used in deep learning models, and applies residual blocks as the base building blocks in their structures. The bottleneck residual block [58] is a variant of the residual block that uses 1×1 convolutions to create a bottleneck. The use of bottlenecks reduces the number of parameters and matrix multiplications. The residual blocks must be made as thin as possible to increase the depth and reduce the number of parameters. The ResNet-50 was adopted as the main structure to extract high-dimensional features and comprises four groups containing three, four, six, and three bottlenecks. The stride of the 3×3 convolutional layer in the second group was two. The remaining three groups did not show changes in the image size.

The feature extraction stage starts with a convolution layer with a kernel size of 3×3 and a stride of two. This stage was followed by batch normalization (BN) and a rectified linear unit (RELU) activation function. After the convolution layer for downsampling, a max-pooling layer with a kernel size of 2×2 was added. Then, ResNet-50 was added for high-dimensional feature extraction, followed by a convolution layer with a kernel size of 3×3 and a stride of two. In the process of feature extraction, an improved ASPP module and DA-ASPP were designed to extract multiscale information [see

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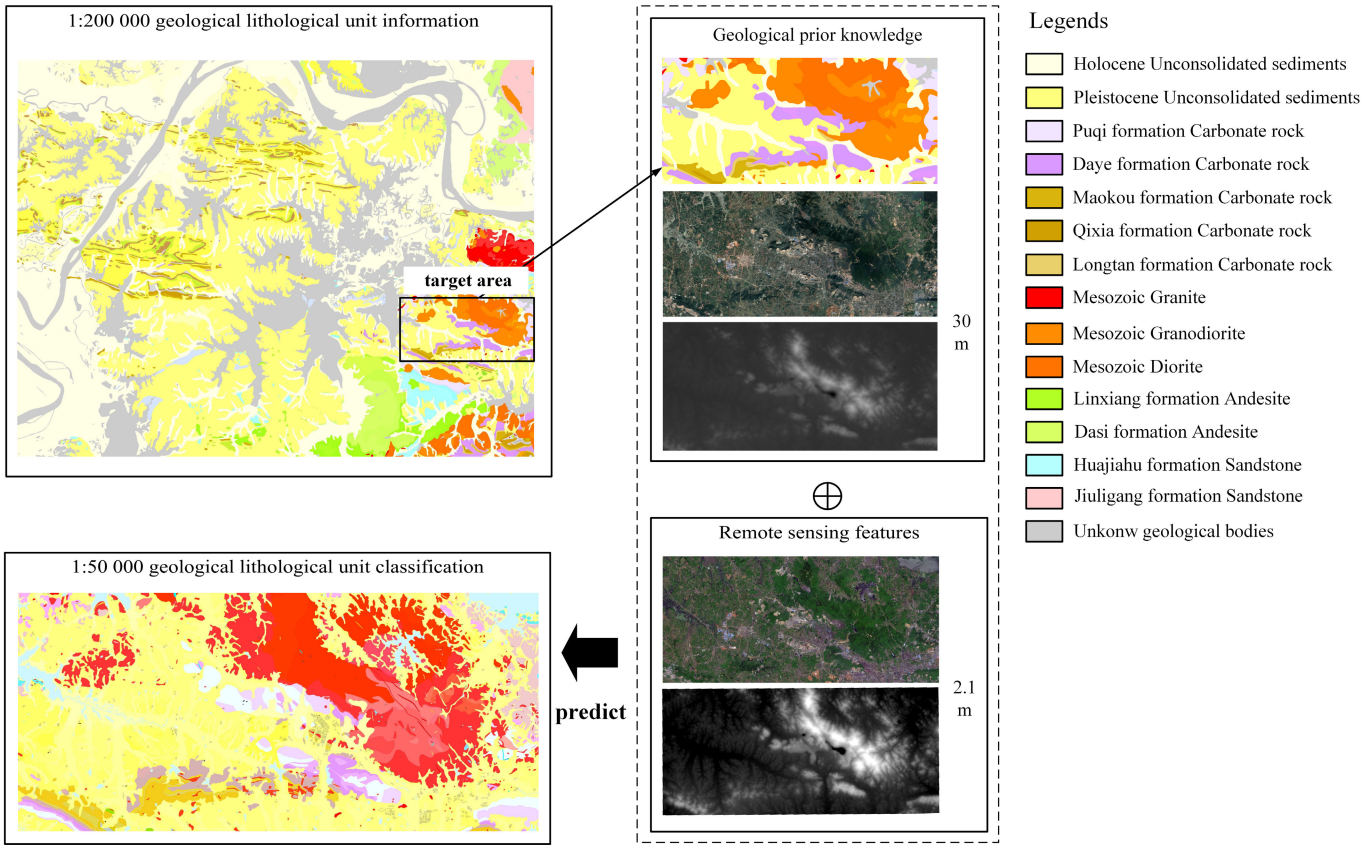


Fig. 1. Prior knowledge and remote sensing data.

Fig. 2(b)] when the size of the image changes (downsampling). A total of four DA-ASPP modules were added to the feature extraction stage.

In DA-ASPP, the concept of dense connection [82] was applied, and four convolution filters were used to extract different-scale features. It was assumed that the input tensor $X_{\text{in}} \in \mathbb{R}^{C\times H\times W}$, where C is the feature dimension, H is the height, and W is the width. First, a convolution layer with a kernel size of 1×1 extracted features from X_{in}. The input of the second, third, and fourth convolution layers was a feature that concatenated the input and results of the previous convolution layer. The kernel sizes of the second, third, and fourth convolution filters were 3×3, and the dilation rates were three, six, and nine, respectively. Then, the results of each convolution layer were merged with X in the channel dimension to obtain the multiscale feature $X' \in \mathbb{R}^{C'\times H\times W}$, $C' = 5C$. Next, in attention residual, channel attention was applied to focus on the important channels in X'. Through global pooling, the features of each channel were compressed to a single pixel ($1 \times 1 \times C'$), the weight matrix was learned through the full connection layer, and the sigmoid function controlled the weight between 0 and 1. Next, the channel attention features were obtained by multiplying the weight matrix with X'. Because the residual structure was easier to train and can avoid the problem of the original features being lost under the attention mechanism, the channel attention features were regarded as residuals and added to X'. Finally, a convolution layer with a kernel size of 1×1 was used to reduce the dimensions and obtain $X_{\text{out}} \in \mathbb{R}^{C\times H\times W}$, which was the output of the feature extraction stage.

2) Feature Projection: As the features extracted from the two branches are from different dimensions, direct fusion may lead to features biased toward one branch. Hence, after obtaining the features from the two branches, prior knowledge and remote sensing features were projected to a new dimension via three convolution layers with a kernel size of 3×3; a BN and RELU activation function followed by each layer.

3) Feature Fusion: Attention residual was applied to the two features after the projection stage, and the results were merged in the channel dimension to fuse these features better.

4) Upsample: Four transposed convolution layers were used to upsample the fused features to the original size of the input image. Finally, a sigmoid classifier was used to classify the lithological units.

C. Study Area and Dataset

The study area is located in Tieshan (see Fig. 3), Edong District, Hubei Province, China, and covers an area of 267.14 km^2 (114°45′ 00″ E–115°00′ 00″ E and 30°10′ 00″ N–30°16′ 00″ N). Edong District is in the westernmost part of the Middle-Lower Yangtze River Metallogenic Belt, which has been one of the most important Cu–Fe–Au–Mo-producing regions of China over the past three decades. This area encompasses over 800 km^2 of igneous rocks and more than 70 deposits, such as porphyry–skarn and skarn Cu–Fe–Au–(Mo–W), related to Jurassic-Cretaceous plutons [83], [84].

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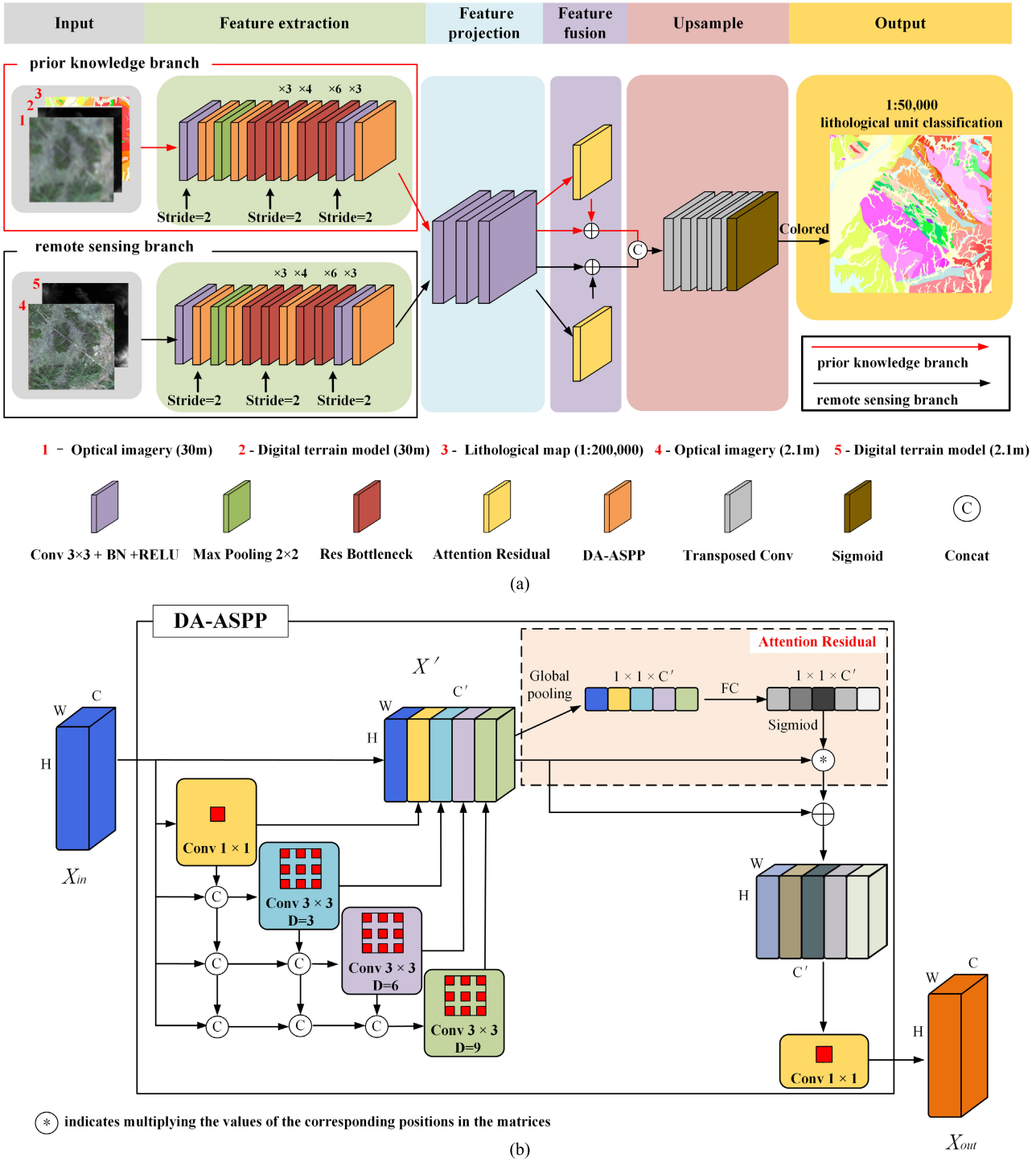


Fig. 2. Framework of dual-branch prior knowledge model (DPNet). (a) Flowchart of the model. (b) Structure of dense attention-atrous spatial pyramid pooling (DA-ASPP). BN: batch normalization; RELU: rectified linear unit.

Because some lithological units had the same properties in the 1:50 000 geological lithological unit information, according to the suggestion by geological experts, we merged them into seven categories: unconsolidated sediments, sandstone, carbonate rock, granite, diorite, water, and mafic dykes (see Fig. 4).

We collected two remote sensing images of different resolutions from two different satellites.

A ZiYuan-3 (ZY-3) stereo satellite image was acquired in 2021. This satellite was equipped with four cameras: one 2.1-m nadir-looking panchromatic camera, two 3.6-m front- and backward-looking panchromatic cameras, and one 5.8-m nadir-looking multispectral camera. First, the 3.6-m resolution front- and backward-looking panchromatic data were used to extract the relative DTM data with a resolution of 10 m using the Environment for Visualizing Images (ENVI) 5.0; the DTM

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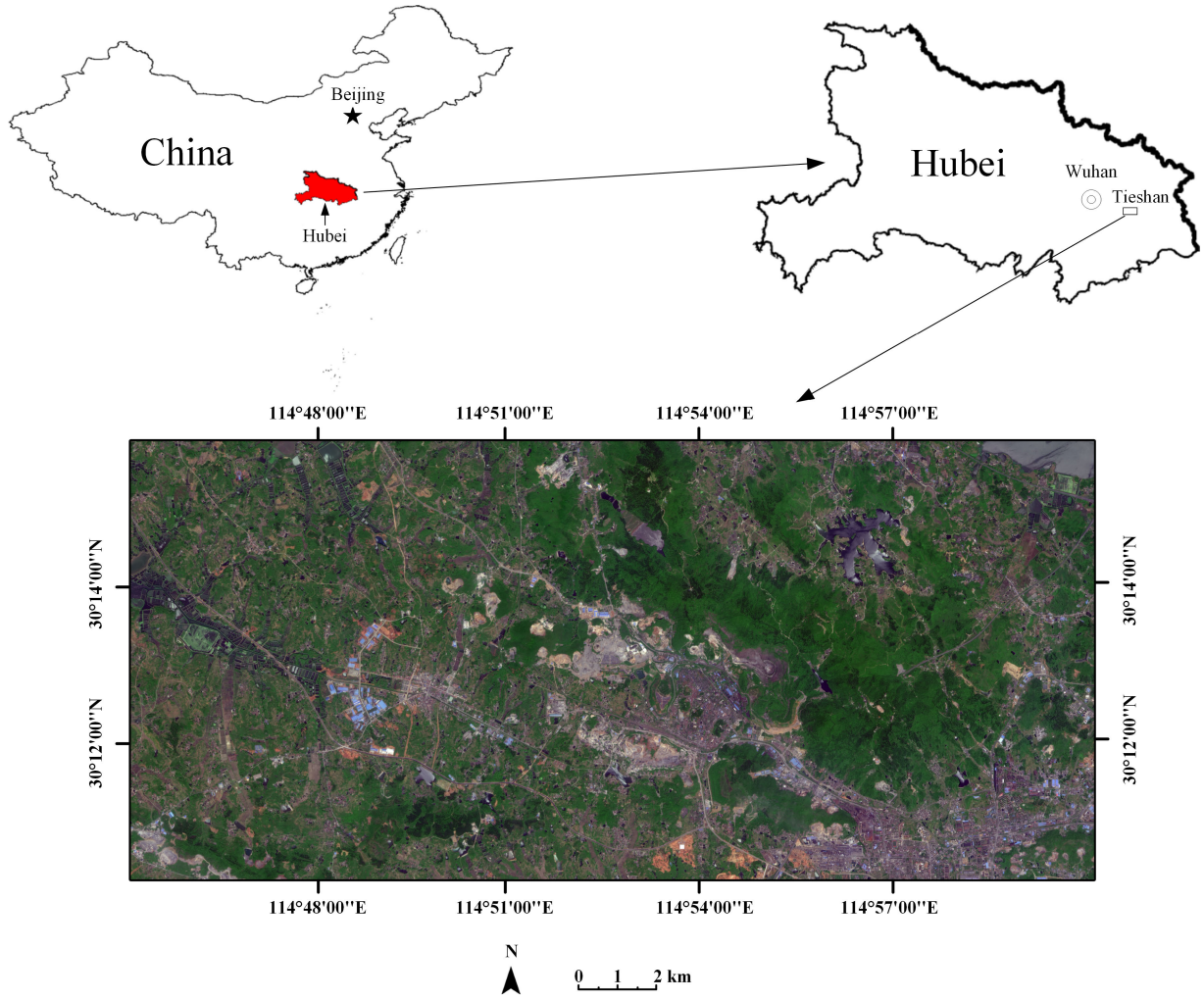


Fig. 3. Map depicting the study area used in this experiment.

was then resampled to 2.1 m. After that, 2.1-m resolution panchromatic-multispectral fused data were generated.

A Landsate-8 satellite image was acquired in 2020, and the multispectral (30 m) and panchromatic (15 m) bands were fused in ENVI to obtain 15 m of multispectral data. Then, DTM data with a resolution of 30 m were downloaded from NASA Earth data. Thereafter, 30-m resolution panchromatic-multispectral fused data were generated.

The 2.1-m resolution panchromatic-multispectral fused data was used as the input features of the remote sensing branch. While the 30-m resolution panchromatic-multispectral fused data combined with 1:200 000 geological lithological unit information was used as the input features of the prior knowledge branch.

Fig. 5 shows the original 1:200 000 geological lithological unit information of the study area, which, we transformed to the lithological unit classes of 1:50 000. Compared with 1:50 000 geological lithological unit information (see Fig. 4), the 1:200 000 geological lithological unit information was coarser, and the corresponding rate between them was 75.48%. Therefore, the accuracy of the deep learning model needs to be better than 75.48%, otherwise, there is no need to make a prediction. The detailed comparison results can be found in Table I.

In order to train deep learning models, the study area was divided into 45 areas in longitude and 22 areas in latitude according to a patch size of 256 × 256 pixels. There was an overlap of 84.4% between zones T and U, and an overlap of 75% between zones 44 and 45 (see Fig. 6).

The training, test, and validation sets were randomly sampled from areas A–U, and the three datasets were ensured to have samples in each area to ensure that the data were evenly distributed in space. Moreover, to maintain consistency in the class distribution between the datasets, it was fine-tuned manually following random sampling to achieve a distribution balance among the classes in the training, test, and validation sets. Finally, there were 567 images in the training set and 189 in the test and validation sets at a ratio of 6:2:2. Fig. 6 depicts the dataset distribution in the study area, and Table II provides details of the dataset.

All experiments were repeated five times, and the mean value and standard deviation were used for comparison. The environment and framework used were Python v3.6 and PyTorch 1.1.0, respectively. The hardware configuration had 64 GB of memory, and the CPU was an Intel(R) Xeon(R)

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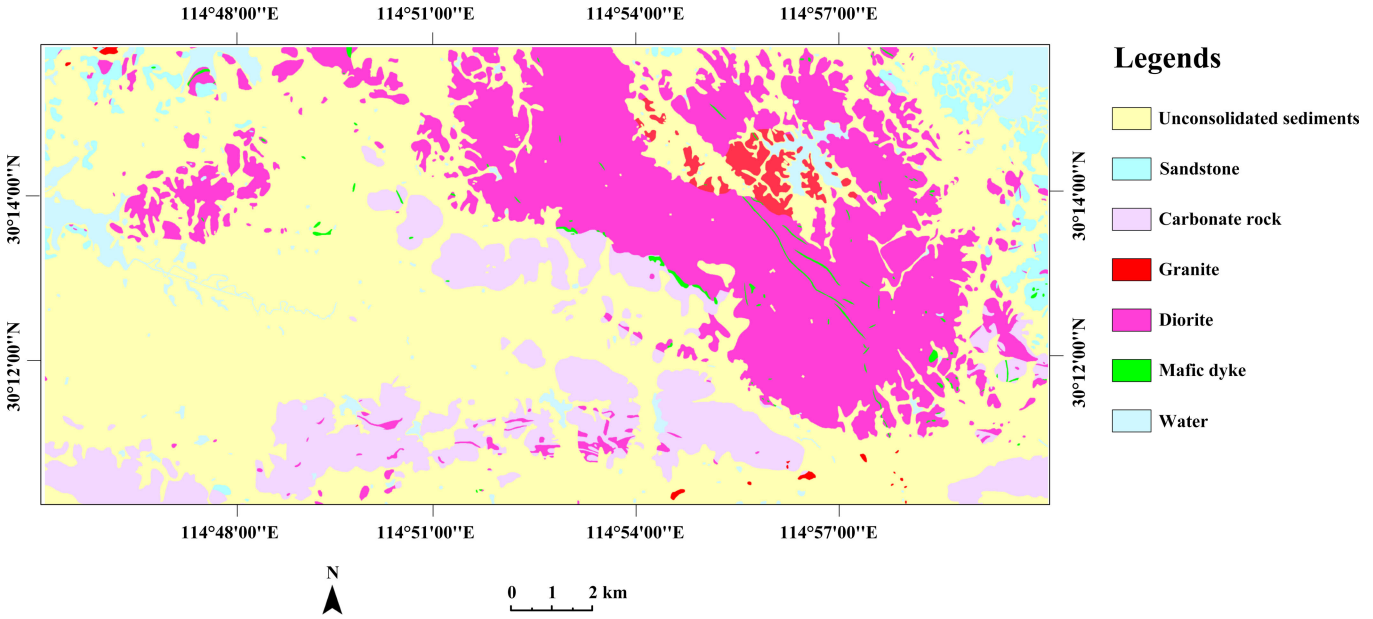


Fig. 4. Lithological map of the Tieshan area, Hubei, China, depicting the composition of the land.

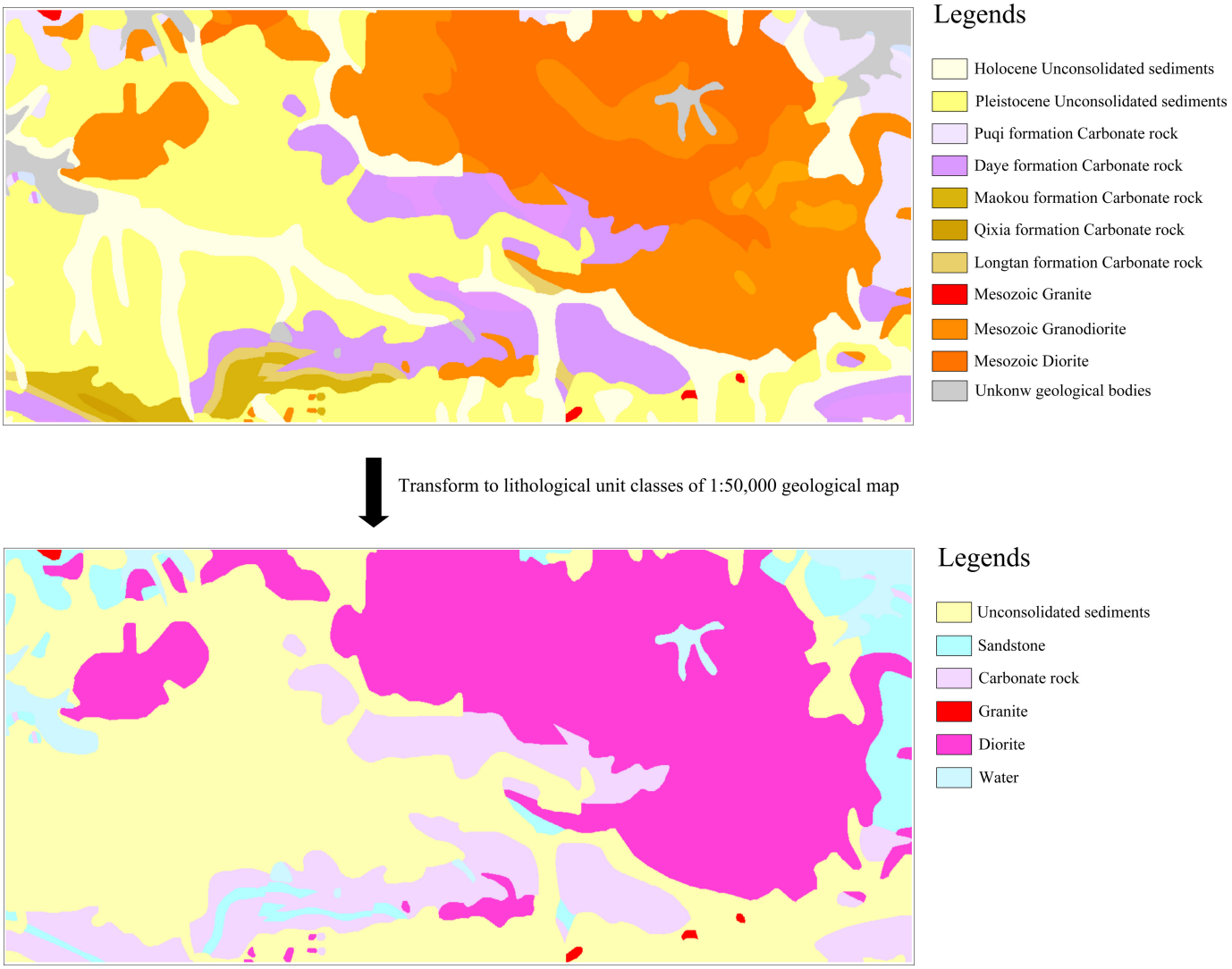


Fig. 5. Geological lithological unit of 1:200 000 and the transformed version.

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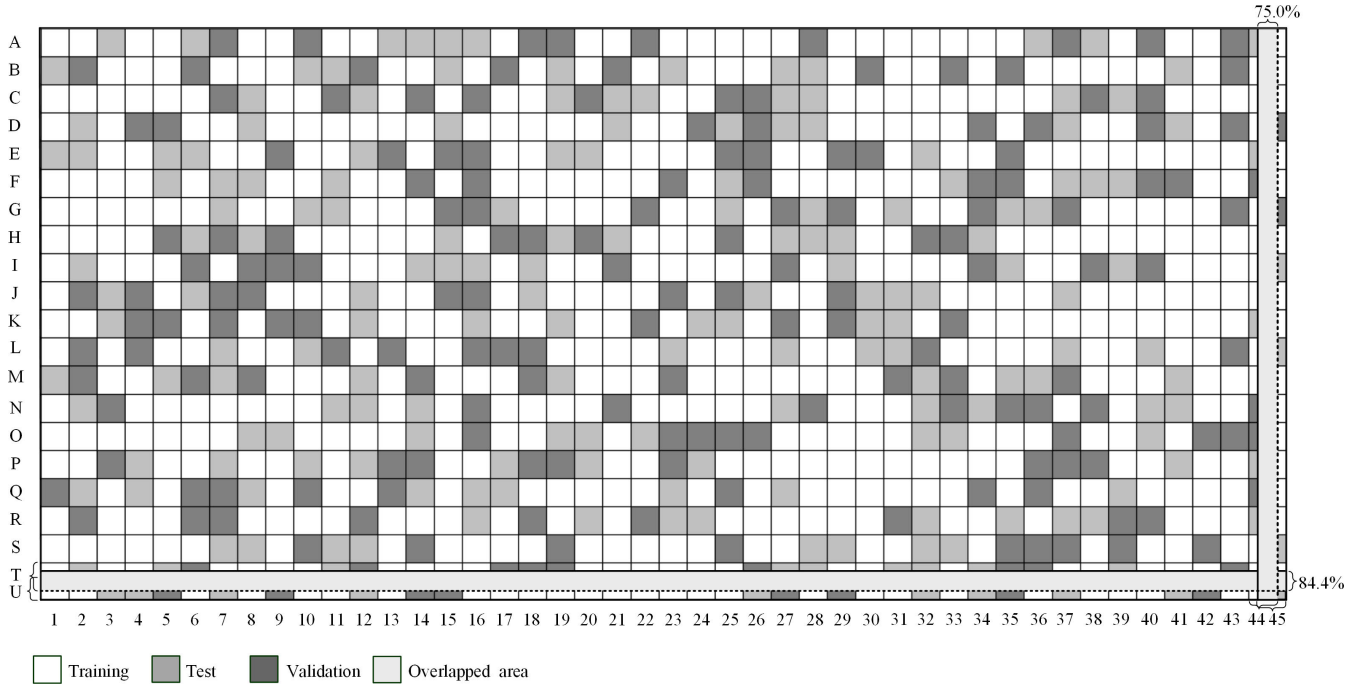


Fig. 6. Division of the study area into 45 areas in longitude and 22 areas (from A to U) and dataset distribution.

TABLE I
QUANTITATIVE COMPARISON OF 1:200 000 AND 1:50 000 GEOLOGICAL LITHOLOGICAL UNIT INFORMATION USING 1:50 000 AS THE LABEL

	Unconsolidated sediments	**Sandstone**	**Carbonate rock**	**Granite**	**Diorite**	**Mafic dyke**	**Water**	**Mean Value**
IOU (%)	67.19	25.20	57.16	4.58	61.98	0	45.39	37.36
F1-score (%)	80.38	40.26	72.74	8.75	76.53	0	62.44	48.73
OA (%)				75.48				

IOU: intersection over union; OA: overall accuracy

TABLE II
CLASS DISTRIBUTIONS IN TRAINING, TEST, AND VALIDATION SETS

No.	Lithological units	Class distribution (%)		
		Training	Test	Validation
1	Unconsolidated sediments	56.93	56.43	58.04
2	Sandstone	2.66	2.03	1.79
3	Carbonate rock	12.57	12.37	11.67
4	Granite	0.94	1.34	0.83
5	Diorite	23.00	25.11	23.22
6	Mafic dyke	0.34	0.40	0.31
7	Water	3.57	2.33	4.14

Silver 4210 and RTX A5000 GPU with 24 GB of memory. The batch size was eight. Furthermore, cross-entropy was used as the loss function, whereas RELU was used as the activation function.

D. Evaluation Metrics

A pixel-based semantic segmentation task was implemented in the experiments. DPNet and other mainstream deep learning models were evaluated using the overall accuracy (OA), F1-score, intersection over union (IOU), mean IOU (mIOU, average value of IOUs), Macro-F1 (average value of F1-scores), floating-point operations (FLOPs), and the number of parameters (Params). The evaluation metrics were defined as follows:

$$\text{Overall accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{1}$$

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TABLE III
QUANTITATIVE COMPARISON OF DIFFERENT BRANCHES AND MODULES FOR DPNET ON THE TEST DATASET OF THE STUDY AREA

Classes	Without DA-ASPP module		Contain DA-ASPP module					
	Dual branches		Dual branches		Remote sensing branch		Prior knowledge branch	
	IOU	F1	IOU	F1	IOU	F1	IOU	F1
Unconsolidated sediments	75.97 ±0.76	86.34 ±0.49	76.34±0.48	86.58±0.32	73.24 ±0.48	84.55 ±0.32	75.22 ±0.18	85.86 ±0.12
Sandstone	43.85 ±1.64	60.59 ±1.59	48.82±0.44	65.61±0.40	37.66 ±1.53	54.70 ±1.61	43.48 ±3.47	60.55 ±3.42
Carbonate rock	59.44 ±2.01	74.55 ±1.59	65.22±0.60	78.95±0.43	50.38 ±5.02	66.91 ±4.42	62.85 ±2.15	77.17 ±1.63
Granite	2.29 ±1.61	4.45 ±3.08	20.24±3.89	33.54±5.48	8.71 ±2.87	15.94 ±4.79	21.82 ±2.44	35.77 ±3.30
Diorite	67.87 ±0.37	80.86 ±0.26	68.40±0.29	81.24±0.20	61.44 ±1.27	76.11 ±0.98	66.95 ±0.83	80.20 ±0.59
Mafic dyke	0	0	0	0	0	0	0	0
Water	32.00 ±1.07	48.47 ±1.24	33.25±1.09	49.90±1.23	28.74 ±1.47	44.63 ±1.78	31.65 ±1.96	48.06 ±2.28
OA	81.59 ±0.40		82.35±0.02		78.85 ±0.72		81.30 ±0.31	
mIOU	40.20 ±0.19		44.61±0.39		37.17 ±0.51		43.14 ±0.35	
Macro-F1	50.80 ±0.36		56.54±0.61		48.98 ±0.41		55.37 ±0.36	

IOU: intersection over union; OA: overall accuracy; DPNET: dual-branch prior knowledge model; DA-ASPP: improved Atrous Spatial Pyramid Pooling

$$\mathrm{F1-score} = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{2}$$

$$\mathrm{IOU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{3}$$

where true positive (TP) is the number of pixels that are correctly extracted as A when evaluating the accuracy of object A. False positive (FP) is the number of other object pixels that are extracted as A. True negative (TN) is the number of pixels that are correctly extracted as another object. False negative (FN) is the number of A pixels extracted as another object.

III. EXPERIMENTAL RESULTS

The results of the different branches in DPNet were compared, and several comparative experiments were conducted using other deep learning models. The classic semantic segmentation model was most used in a previous study of lithology classification based on deep learning [15], [16]. Although Han et al. [14] made the latest progress in lithology classification in 2022, they did not provide a source code; therefore, their model could not be used for comparison.

Five common semantic segmentation models (UNet, UNet++, Attention UNet, DeepLab v3+, and PSPNet) and two recent models (Swin UNet and Bi-HRNet) were chosen for comparison. The UNet is a classical and commonly used semantic segmentation model. The PSPNet uses a pyramid pooling module to aggregate global context information. DeepLab v3+ uses dilated convolution and ASPP to expand the receptive field and capture contextual information. UNet++ focuses on multiscale features. The attention UNet uses an attention mechanism to weaken background features. The recent models (Swin UNet and Bi-HRNet) achieved state-of-the-art results in computer vision.

A. Comparison of Different Branches and Modules

DPNet was examined with different branches and modules to verify the importance of prior knowledge and the DA-ASPP module.

Table III lists the evaluation metrics for the results of DPNet with different modules and branches using the test set in the study area. Compared with directly transforming 1:200 000 geological lithological unit information to 1:50 000 geological lithological unit information, which has an OA of 75.48%, mIOU of 37.36%, and Macro-F1 of 48.73% (see Table I for details), DPNet achieved 82.35% in OA, 44.61% in mIOU, and 56.54% in Macro-F1. These results indicate that, compared with 1:200 000 geological lithological unit information, DPNet can provide a more accurate reference in the absence of 1:50 000 geological lithological unit information.

When using different branches in DPNet, it is obvious that prior knowledge branches have an important impact on the accuracy of the LUC. After adding prior knowledge, the value of each evaluation metric (OA, mIOU, and Macro-F1) increased by approximately 5%. For most lithologic units (sandstone, carbonate rock, granite, diorite, and water), the IOU and F1-score values increased by approximately 5%–15%. Due to the small sample number of the mafic dykes in the study area, both strategies failed to classify it correctly. The results show that the dual-branch strategy is superior to the single-branch strategy. Compared to using the prior knowledge branch only, the values of OA, mIOU, and Macro-F1 increased by approximately 1% in the dual-branch strategy.

When verifying the effectiveness of the DA-ASPP, the results obtained without the DA-ASPP module were relatively poor. However, the addition of DA-ASPP improved the classification accuracy of each type of lithologic unit (except mafic

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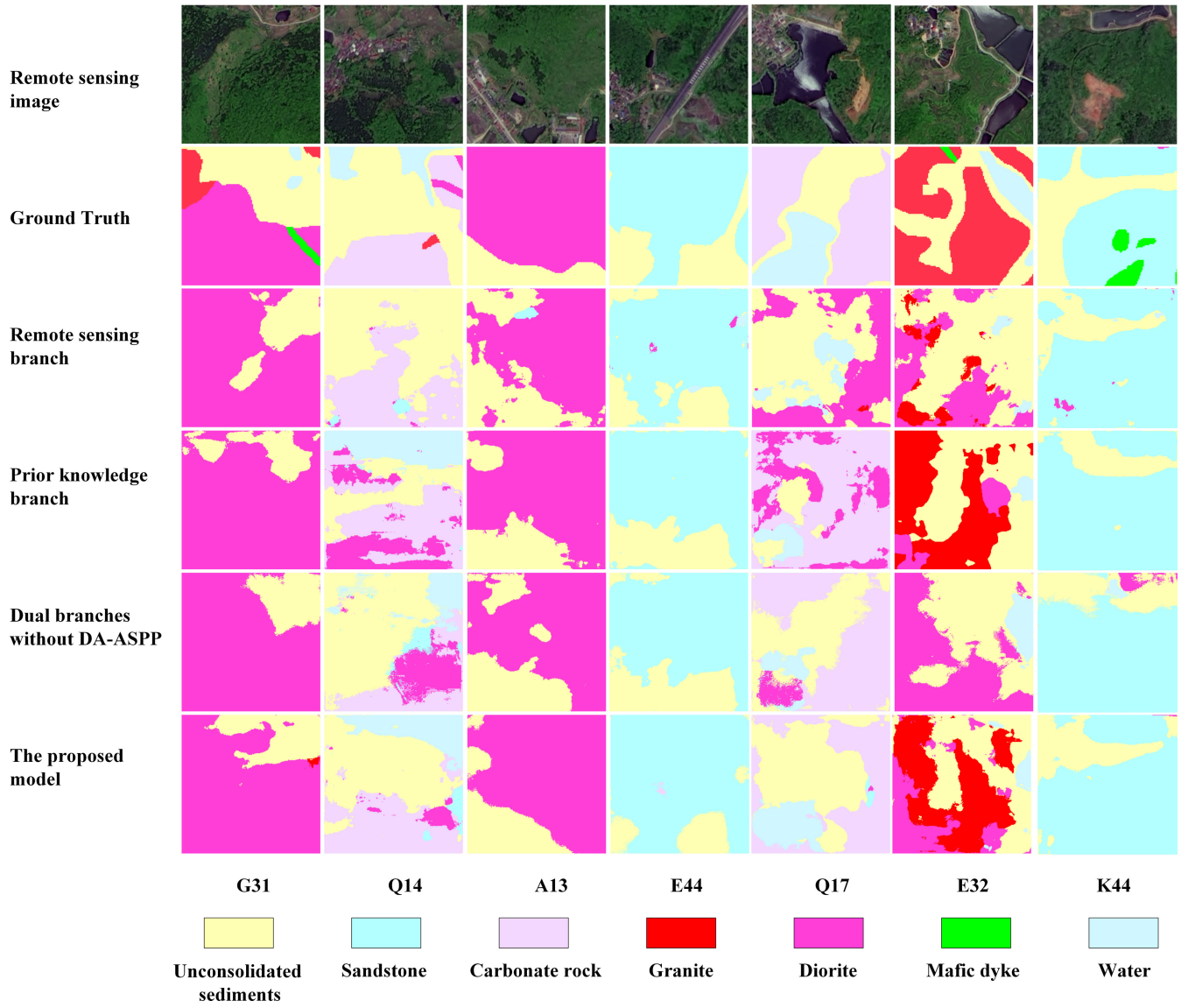


Fig. 7. Visual comparison of different branches and modules using representative images in the study area.

dyke), and for OA, mIOU, and Macro-F1, it increased by approximately 1%. Therefore, the results in Table III verify that both the prior knowledge branch and the DA-ASPP module in DPNet are beneficial.

Fig. 7 depicts the performance of DPNet with different branches and modules. Seven representative images were used for visual comparison. Areas G31 and Q14 contain five lithologic units (diorite, granite, water, unconsolidated sediments, and mafic dykes in G31, and unconsolidated sediments, carbonate rock, diorite, granite, and water in Q14) to compare the performance of the model in complex scenes. The rest of the images, in which one or two types of lithologic units cover a large area, were mainly used to compare the performance of the model in classifying each lithologic unit. A13 mainly consists of diorite, E44 is sandstone and unconsolidated sediments, and Q17 is carbonate rock and water. Because granite and mafic dykes account for a small proportion of the study area, areas E32 and K44 were selected, which are relatively obvious in these two lithologic units.

Because of the vegetation cover (see Fig. 7), when applying remote sensing branches only, other lithologic units are often mistakenly classified as unconsolidated sediments (G31, A13, E44, H17, and K44). After combining the prior knowledge branch and the DA-ASPP module, the classification accuracy of most lithologic units improved, such as carbonate rock in Q14, sandstone in E44, water in Q17, and granite in E32. However, when classifying lithologic units in complex scenes (Q14), although DPNet performed better, it was still inaccurate in identifying lithologic unit boundaries.

B. Comparison of Different Models

Seven classical and recent models were compared with DPNet. The input features of prior knowledge branch and remote sensing branch were combined in the channel dimension, and used as the inputs for these comparison models.

Table IV summarizes the evaluation metrics for the results of DPNet and comparison models when using the test set in

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TABLE IV
PERFORMANCE COMPARISONS OF DIFFERENT MODELS ON THE TEST DATASET OF THE STUDY AREA

Classes	UNet		UNet++		Attention UNet		DeepLab v3+		PSPNet		Bi-HRNet		SwinUNet		DPNet	
	IOU	F1	IOU	F1	IOU	F1	IOU	F1	IOU	F1	IOU	F1	IOU	F1	IOU	F1
Unconsolidated	76.79	86.87	75.67	86.15	76.06	86.40	74.28	85.24	75.70	86.16	72.93	84.34	75.97	86.34	76.34	86.58
sediments	±0.31	±0.36	±0.31	±0.20	±0.40	±0.26	±0.28	±0.19	±0.79	±0.52	±0.84	±0.57	±0.76	±0.49	±0.48	±0.32
Sandstone	42.62	59.69	30.22	46.08	30.11	45.97	43.73	60.73	43.86	60.95	41.56	58.70	43.85	60.95	48.82	65.61
	±3.97	±3.96	±7.40	±8.64	±7.27	±8.47	±5.22	±4.95	±2.36	±2.27	±1.99	±1.99	±1.64	±1.59	±0.44	±0.40
Carbonate rock	63.00	77.30	58.79	74.04	60.04	75.03	62.35	76.80	63.57	77.71	61.88	76.45	59.44	74.55	65.22	78.95
	±0.37	±0.28	±0.84	±0.67	±1.07	±0.83	±1.45	±1.11	±2.49	±1.88	±0.23	±0.17	±2.01	±1.59	±0.60	±0.43
Granite	3.92	7.37	1.84	3.49	0.13	0.25	5.94	10.92	18.39	30.26	3.30	6.26	2.29	4.45	20.24	33.54
	±3.75	±6.97	±3.18	±6.02	±0.22	±0.43	±5.09	±8.89	±9.86	±14.55	±3.22	±6.05	±1.61	±3.08	±3.89	±5.48
Diorite	67.28	80.44	67.90	88.08	68.10	81.02	65.56	79.19	67.44	80.56	63.48	77.66	67.87	80.86	68.40	81.24
	±0.51	±0.37	±0.34	±0.24	±0.31	±0.22	±0.39	±0.28	±0.13	±0.09	±0.64	±0.48	±0.37	±0.26	±0.29	±0.20
Mafic dyke	0	0	0	0	0	0	2.71	5.21	0	0	4.89	9.26	0	0	0	0
							±2.36	±4.53			±2.22	±4.07				
Water	32.15	48.62	33.13	49.75	33.77	50.48	26.71	42.11	31.51	47.84	26.28	41.62	32.00	48.47	33.25	49.90
	±2.53	±2.88	±1.68	±1.88	±0.62	±0.68	±2.72	±3.36	±3.54	±4.11	±0.85	±1.08	±1.07	±1.24	±1.09	±1.23
OA	81.96±0.31		81.30±0.12		81.49±0.22		80.66±0.20		81.68±0.58		79.54±0.47		79.31±0.23		82.35±0.02	
mIOU	40.82±0.71		38.22±0.86		38.32±1.06		40.18±0.80		42.92±1.26		39.19±0.47		40.20±0.19		44.61±0.39	
Macro-F1	51.47±1.16		48.63±1.12		48.45±1.24		51.46±0.63		54.78±2.10		50.61±0.63		50.80±0.36		56.54±0.61	
FLOPs	21.02G		42.64G		21.42G		8.55G		19.17G		8.75G		11.92G		49.83G	
Param	31.04M		34.92M		31.39M		59.35M		48.70M		29.51M		27.17M		102.27M	

UNet; PSPNet: Pyramid Scene Parsing Network; Bi-HRNet: High resolution network; SwimUNet; DPNet: dual-branch prior knowledge model.

the study area. Among these models, DPNet exhibited the best performance, with an OA of 82.35%, mIOU of 44.61%, and Macro-F1 of 56.54%. Compared with the second-best model (PSPNet), OA was 0.6% higher, and mIOU and Macro-F1 were approximately 1.5% higher.

For the classification of each lithologic unit, DPNet showed the best IOU and F1-score for sandstone, carbonate rock, diorite, and granite. In particular, sandstone and granite exceeded 2% of the evaluation metrics in all comparison models. For the remaining lithologic unit categories (except mafic dykes), the accuracy difference between the best model and DPNet was within only 0.5%. The results demonstrate the effectiveness of DPNet. However, it is noteworthy that only DeepLab v3+ and Bi-HRNet can correctly classify mafic dykes.

Fig. 8 depicts the performance of the eight methods, as outlined in Table IV. Again, the representative images in Fig. 7 were used for visual comparison. DPNet was better at classifying lithologic units in complex scenes. In area Q14, DPNet classified the carbonate rock more accurately, and the classification results were closer to the ground truth. In area G31, DPNet, UNet++, and PSPNet distinguished diorite better, but no model could classify other lithologic units in the scene correctly. For scenes with a single lithology category (areas A13, E44, Q17, and E32), PSPNet and DPNet performed better. In Q17 and E32, their classification results were closer to the ground truth, such as water in Q17 and granite in E32. However, only DeepLab v3+ and Bi-HRNet can identify mafic dykes in E44. Furthermore, the boundary of the lithologic unit divided by DeepLab v3+ was smoother. From the overall visualization results, the model based on the SPP structure (DeepLab v3+, PSPNet, and the model proposed by this study) performed better, which is also reflected in the lithological interpretation research of Han [14].

IV. DISCUSSION

A. Effects of the Different Modules and Branches Combinations and the Innovation of the Proposed Module

The results of the prior knowledge branch were considerably better than those of the remote sensing branch when comparing the effectiveness of each branch. Because the network structures of these two branches are the same, the classification accuracy of each lithologic unit was improved (sandstone, carbonate rock, granite, and diorite increased by >5%) after adding prior knowledge (1:200 000 geological lithological unit information). This improvement shows that prior knowledge can improve the accuracy of LUC. Furthermore, the visualization results can illustrate this improvement, especially granite in E32. And after adding the DA-ASPP module, the classification accuracy of water (Q17 in Fig. 7) and granite (E32 in Fig. 7) was substantially better. It is worth noting that the results of the prior knowledge branch were better when comparing it to the DA-ASPP and to the double branch without the DA-ASPP. This result shows that the designed DA-ASPP module can use prior knowledge better.

The application of prior knowledge branch was innovative. Due to vegetation cover, it is not easy to distinguish lithologic units based solely on remote sensing images. Therefore,

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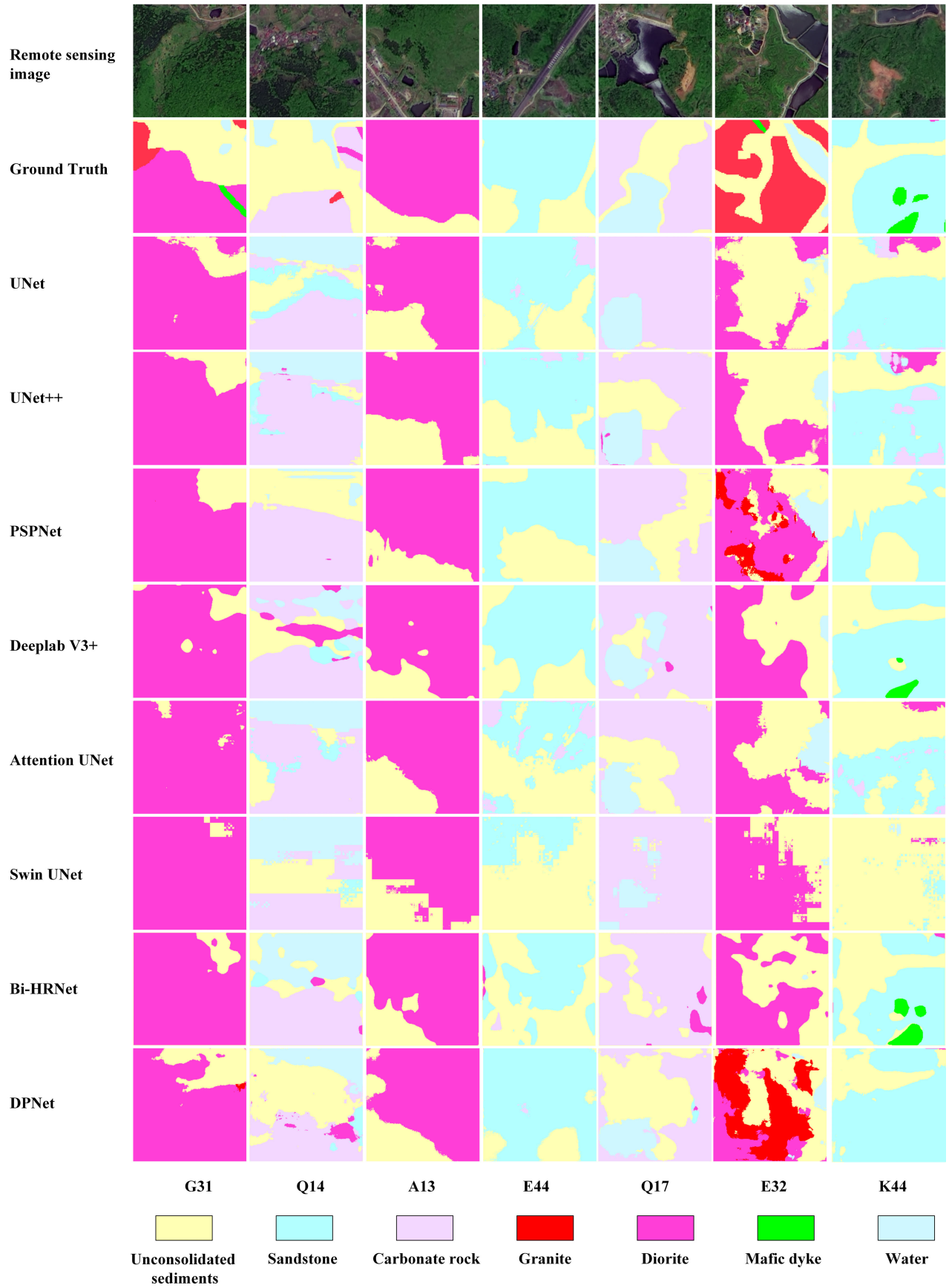


Fig. 8. Visual comparison of different models using representative images in the study area.

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geological knowledge must be added to help identify the lithologic units. Because small-scale geological lithological units contain a lot of geological background information, it was added into the model as prior knowledge in large-scale geological interpretations. And this research is the first to utilize deep learning features and prior geological knowledge for LUC.

The designed DA-ASPP module was innovative. ASPP can improve the receptive field of the model during the feature extraction section. Therefore, the extracted features of each receptive field were densely connected, and make the extracted features contain multiscale information of lithologic units. In addition, the application of channel attention module can screen out the representative features of lithologic units. Simultaneously, the residual structure can prevent the occurrence of degradation.

B. Effects of the Model and Comparison Models

The model with the SPP module, which is designed to extract multiscale information, performs better on mIOU (Deeplab v3+, PSPNet, and our model). DeepLab v3+ adds dilated convolution in SPP and applies the Xception structure to obtain multiresolution information. It shows better results when classifying mafic dykes. Similarly, Bi-HRNet, which uses a dense connection structure to extract multiresolution information, can also classify mafic dykes. Hence, for lithologic units, which are long strips with a small number of samples, multiresolution feature extraction of input information may be beneficial for retaining their key information. Owing to the limitation of the receptive field, the results of UNet, UNet++, and Attention UNet are not very good. Finally, for Swin UNet, although the Swin transformer structure has a long-distance receptive field and pixel-level self-attention mechanism, its performance is poor. Because it is proposed to segment medical images, we believe that this structure (Swin UNet) is unsuitable for remote sensing or geological interpretation, as their input features are relatively complex.

C. Limitations and Scope for Improvement

DPNet had several limitations. The first limitation is that although prior knowledge can improve the classification accuracy of lithologic units in vegetation coverage areas (see Fig. 7), it is still difficult to accurately identify the boundaries of lithologic units when using remote sensing images for lithologic classification, and there are some noise points in the interpreted boundaries. For example, in E44 of Fig. 8, the classification of sandstone contains some noise points of carbonate rock and vice versa in Q14 of Fig. 8. The second limitation is that the classification accuracy of DPNet for a mafic dyke was 0%. In addition, the accuracy of the lithologic classification of the model was low for complex scenes. For example, in E44 of Fig. 8, DPNet could not identify the mafic dyke. And in G31 of Fig. 8, which is a complex scene and contains six types of lithological units, DPNet only classify two of them.

The proposed dataset had several limitations. First, most small-scale geological surveys were produced in the 1980s, whereas remote sensing images were produced in recent years. Although most rocks do not change, the position of the water may deviate. Therefore, this position must be corrected according to the remote sensing image. Second, although China has carried out large-scale geological mapping in some regions, obtaining this data is difficult, which may limit the regional generalization ability of the model. Moreover, the number of samples of certain classes was minimal, affecting the sample distribution substantially. Third, according to previous geological and lithologic interpretation studies, synthetic aperture radar (SAR), geochemical, and geophysical data also apply to LUC; however, the dataset in this study lacked this information. Finally, the lithology of some areas was inferred when conducting geological mapping, this may not be particularly accurate as a label. Therefore, the geological lithological unit information can only be used as a reference, requiring further field investigations.

Several targeted improvements should be implemented to address these limitations, the first being model improvement. For example, the dense connections between multifeature fusion and multiresolution information may improve the classification accuracy in the study area. Furthermore, a geological knowledge graph can be used to learn prior knowledge. The second improvement should be dataset improvement. Considering the limitations of the proposed dataset, the research area should be expanded; SAR and geochemical data should be added to enrich the dataset. In addition, field survey points should be set up, and labels should be corrected.

V. Conclusion

Geological lithological unit information is important for resource reconnaissance, infrastructure planning, and landslide/environmental governance. However, vegetation coverage makes it difficult to distinguish lithologic units based solely on remote sensing images. Therefore, this study proposes the combination of prior geological knowledge and deep features of remote sensing images to create a two-branch deep learning model to interpret geological lithologic units. A prior knowledge branch was designed to obtain geological background information, and a remote sensing image branch was applied to extract remote sensing deep features. ResNet-50 was used as the backbone of each branch. In addition, a DA-ASPP module that integrates the idea of dense connection into the ASPP and adds a residual structure into the channel attention mechanism was constructed to extract key information.

Several conclusions can be drawn by comparing several classical and recent models to the model in this study. First, the prior knowledge branch and DA-ASPP modules help classify lithologic units, and the value of mIOU and Macro-F1 can be increased by approximately 5% with these two strategies. Second, DPNet performs better than other comparable models. Further, this study included several innovations. The first is using small-scale geological lithological unit information as prior geological knowledge to provide geological background information. The second innovation was proposing a dual-branch deep learning model with an improved ASPP

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module for multiscale feature extraction and screening out of the representative features of lithologic units. Because the proposed method is effective, a preliminary result can be quickly obtained for reference, especially in areas where field surveys are difficult to carry out. However, there are still issues with accurate lithologic boundary identification. Therefore, the effects of the dense connections of multifeature fusion and multiresolution information should be evaluated to enhance the ability to classify each category.

Model and dataset improvement are key points in future work. For model improvement, we will try to use morphological methods to eliminate the noise of boundaries, and then use feature enhancement methods to improve the classification accuracy. For dataset improvement, multidata, such as SAR, geochemical, and geophysical data, will be added in our dataset to provide more information.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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